UAV System Model

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Fig. 1. UAV-assisted emergency communication network with ground users, hazardous zones, and the UAV trajectory from the initial to the final position.

I. Introduction

Notations: In this paper, scalars, vectors, and matrices are denoted by lowercase letters x, bold lowercase letters $\mathbf{x}$, and bold uppercase letters $\mathbf{X}$, respectively. The length of a vector $\mathbf{x}$ is denoted by $|\mathbf{x}|$. The spaces of $M \times N$ complex-valued and real-valued vectors are denoted by $\mathbb{C}^{M\times N}$ and $\mathbb{R}^{M\times N}$, respectively. The notation $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ indicates that $\mathbf{x}$ follows a complex Gaussian distribution with mean vector $\boldsymbol{\mu}$ and covariance matrix σ^2, whereas $\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \sigma^2)$ denotes a real normal distribution. An N-dimensional all-ones vector is denoted by $\mathbf{1}_N$. The symbols $\mathbf{X}^T$ and $\mathbf{X}^H$ represent the transpose and the conjugate transpose of $\mathbf{X}$, respectively. Furthermore, $\|\cdot\|$ denotes the Euclidean norm, and $\otimes$ represents the tensor product between quantum states. Throughout this paper, these notations are adopted consistently for clarity.

II. System Model

In this project, we consider a UAV-assisted emergency communication network deployed over a crisis-affected area, such as a region struck by a natural disaster or a conflict zone, where the terrestrial communication infrastructure has been destroyed or is unavailable, as illustrated in Fig. 1. A rotary-wing UAV equipped with a single antenna acts as an aerial base station and delivers downlink rescue information (e.g., evacuation routes, medical instructions, and situational updates) to a set of M single-antenna ground users, denoted by $\mathcal{M} = \{1, 2, \ldots, M\}$.

A. Network Model

The ground users are randomly located within a two-dimensional square region of dimensions $D_0 \times D_0$. The horizontal coordinates of the m-th user are represented by $\boldsymbol{\psi}_m = [\psi_m^x, \psi_m^y, 0]$, which are assumed to be known in advance at the UAV, e.g., from emergency calls or prior surveys. The mission duration T is divided into N equal time slots of length $\delta = T/N$, indexed by $\mathcal{N} = \{1, 2, \ldots, N\}$. The slot length δ is chosen sufficiently small such that the UAV position can be regarded as constant within each slot.

The UAV flies at a fixed altitude H_u, and its horizontal position in time slot n is denoted by $\boldsymbol{\psi}_u[n] = [\psi_u^x[n], \psi_u^y[n]]$. Accordingly, the three-dimensional (3D) Euclidean distance between user m and the UAV in time slot n is given by

$$d_m[n] = \sqrt{(\psi_m^x - \psi_u^x[n])^2 + (\psi_m^y - \psi_u^y[n])^2 + H_u^2}. \quad (1)$$

Furthermore, the area contains a set of J hazardous zones, denoted by $\mathcal{J} = \{1, 2, \ldots, J\}$, such as hostile areas, active fire fronts, or restricted airspace, which the UAV must avoid. The j-th hazardous zone is modeled as a disk with center $\boldsymbol{\psi}_j^{\mathrm{z}} = [\psi_j^{\mathrm{z},x}, \psi_j^{\mathrm{z},y}]$ and radius r_j.

B. Wireless Channel Model

We consider a crisis scenario in which single-antenna ground users receive downlink signals from a single-antenna UAV. The air-to-ground links are assumed to be dominated by the distance-dependent path loss, and the effect of small-scale fading is neglected. Accordingly, the channel power gain between the UAV and the m-th user in time slot n is modeled as

$$\beta_m[n] = \beta_0 d_m^{-\alpha}[n], \quad (2)$$

where β_0 is the reference channel gain at a unit distance and α is the path-loss exponent, which captures the additional attenuation caused by obstacles such as collapsed buildings and debris. Since $d_m[n]$ depends on the UAV position, the channel gain varies with the UAV trajectory. Moreover, since the user locations are known in advance, the UAV can compute the channel gain of every user directly from its own position using (2).

C. User Scheduling and Downlink Transmission

Let $a_m[n] \in \{0, 1\}$ denote the binary scheduling variable, where $a_m[n] = 1$ indicates that user m is served by the UAV in time slot n, and $a_m[n] = 0$ otherwise. The UAV serves exactly one user in each time slot, i.e.,

$$\sum_{m\in\mathcal{M}} a_m[n] = 1, \quad \forall n \in \mathcal{N}. \quad (3)$$

Hence, the scheduled user occupies the entire system bandwidth B and the maximum transmit power $P_{\max}$ of the

UAV, and no inter-user interference occurs. When user m is scheduled in time slot n, its received signal is

$$y_m[n] = \sqrt{P_{\max}\,\beta_m[n]}\,s_m[n] + z_m[n], \tag{4}$$

where $s_m[n]$ is the unit-power data symbol intended for user m, and $z_m[n] \sim \mathcal{CN}(0, \sigma^2)$ is the additive white Gaussian noise with $\sigma^2 = N_0 B$, where N_0 is the noise power spectral density. Accordingly, the received signal-to-noise ratio (SNR) of user m is

$$\gamma_m[n] = \frac{P_{\max}\,\beta_m[n]}{\sigma^2} = \frac{P_{\max}\,\beta_0}{\sigma^2 d_m^{\alpha}[n]}. \tag{5}$$

The achievable data rate of user m in time slot n is then

$$R_m[n] = a_m[n]\,B\log_2\left(1+\gamma_m[n]\right), \tag{6}$$

and the amount of data delivered to user m in time slot n is $\delta R_m[n]$ bits. Since only one user is served in each time slot, maximizing the sum rate alone would always favor serving the user with the strongest channel, typically the one closest to the UAV, which motivates the per-user data demand constraint introduced in the next section.

D. UAV Mobility and Energy Model

The UAV departs from an initial position $\boldsymbol{\psi}_{\mathrm{I}}$ and must arrive at a final position $\boldsymbol{\psi}_{\mathrm{F}}$ (e.g., a recharging or base station) by the end of the mission, i.e.,

$$\boldsymbol{\psi}_u[1] = \boldsymbol{\psi}_{\mathrm{I}}, \qquad \boldsymbol{\psi}_u[N] = \boldsymbol{\psi}_{\mathrm{F}}. \tag{7}$$

Given the maximum UAV speed $V_{\max}$, the trajectory must satisfy

$$\|\boldsymbol{\psi}_u[n+1] - \boldsymbol{\psi}_u[n]\| \le V_{\max}\,\delta, \quad n = 1, \ldots, N-1. \tag{8}$$

Let $V[n] = \|\boldsymbol{\psi}_u[n+1] - \boldsymbol{\psi}_u[n]\|/\delta$ denote the UAV flying speed in time slot n. Following the widely adopted rotary-wing UAV propulsion model, the propulsion power consumption is given by

$$\begin{aligned} P(V) = P_0\left(1+\frac{3V^2}{U_{\mathrm{tip}}^2}\right) + \frac{1}{2}d_0\rho s A V^3 \\ + P_{\mathrm{i}}\left(\sqrt{1+\frac{V^4}{4v_0^4}} - \frac{V^2}{2v_0^2}\right)^{1/2}, \end{aligned} \tag{9}$$

where P_0 and P_{i} denote the blade profile power and the induced power in hovering status, respectively, U_{tip} is the tip speed of the rotor blade, v_0 is the mean rotor induced velocity in hovering, d_0 is the fuselage drag ratio, ρ is the air density, s is the rotor solidity, and A is the rotor disc area. The total propulsion energy consumed during the mission must not exceed the on-board battery capacity $E_{\max}$, i.e.,

$$\delta\sum_{n=1}^{N-1} P\left(V[n]\right) \le E_{\max}. \tag{10}$$

III. Problem Formulation

Our objective is to maximize the sum rate of all users over the mission duration by jointly optimizing the binary user scheduling $\mathbf{A} = \{a_m[n], \forall m, n\}$ and the UAV trajectory $\boldsymbol{\Psi} = \{\boldsymbol{\psi}_u[n], \forall n\}$, while guaranteeing that every user receives its required amount of rescue data D_m. The optimization problem is formulated as

$$\mathcal{P}1: \quad \max_{\mathbf{A},\boldsymbol{\Psi}} \sum_{n\in\mathcal{N}}\sum_{m\in\mathcal{M}} R_m[n] \tag{11a}$$

$$\text{s.t.} \quad a_m[n] \in \{0,1\}, \quad \forall m \in \mathcal{M},\, \forall n \in \mathcal{N}, \tag{11b}$$

$$\sum_{m\in\mathcal{M}} a_m[n] = 1, \quad \forall n \in \mathcal{N}, \tag{11c}$$

$$\delta\sum_{n\in\mathcal{N}} R_m[n] \ge D_m, \quad \forall m \in \mathcal{M}, \tag{11d}$$

$$\|\boldsymbol{\psi}_u[n+1] - \boldsymbol{\psi}_u[n]\| \le V_{\max}\,\delta, \quad n = 1, \ldots, N-1, \tag{11e}$$

$$\boldsymbol{\psi}_u[1] = \boldsymbol{\psi}_{\mathrm{I}}, \quad \boldsymbol{\psi}_u[N] = \boldsymbol{\psi}_{\mathrm{F}}, \tag{11f}$$

$$0 \le \psi_u^x[n],\, \psi_u^y[n] \le D_0, \quad \forall n \in \mathcal{N}, \tag{11g}$$

$$\|\boldsymbol{\psi}_u[n] - \boldsymbol{\psi}_j^z\| \ge r_j, \quad \forall j \in \mathcal{J},\, \forall n \in \mathcal{N}, \tag{11h}$$

$$\delta\sum_{n=1}^{N-1} P\left(V[n]\right) \le E_{\max}, \tag{11i}$$

where constraint (11b) defines the binary scheduling variables, and (11c) ensures that exactly one user is served in each time slot. Constraint (11d) ensures that all M users receive their required rescue data within the mission, which prevents the UAV from serving only the users with the best channels. Constraints (11e)–(11g) capture the UAV mobility limitations, (11h) keeps the UAV outside all hazardous zones, and (11i) represents the UAV battery constraint.